

END-TERM PROJECT-01 REPORT

ON

<TITLE OF THE PROJECT>

B.Tech. [Computer Science and Engineering]

Semester:

Section:

Course Title & Code: (CSE)

Submitted By:

Student Name:

Registration No:

Guided By:

Academic Year



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
FACULTY OF ENGINEERING AND TECHNOLOGY, INSTITUTE OF
TECHNICAL EDUCATION AND RESEARCH
SIKSHA 'O' ANUSANDHAN (DEEMED TO BE) UNIVERSITY
BHUBANESWAR, ODISHA, INDIA

ABSTRACT

This project focuses on the design and development of a simple implemented using the C programming language. The main

Submitted By

Guided By

TABLE OF CONTENTS

Section No.	Topic	Page No.
01	Introduction	1
02	Problem Statement	1
03	Objectives of the Project	1
04	Tools and Technologies Used	2
05	Methodology	3
06	Results Analysis and Screenshots	4
07	Logical Understanding	5
08	Conclusion	6-7
09	Future Scopes and Application	8
	References	8
	Annexure	9-15

LIST OF FIGURES

Figure No.	Figure Title	Page No.
Figure 1	...	6
Figure 2	...	7
Figure 3	...	7

01. Introduction

C programming is one of the most widely used and fundamental programming languages in computer science. It provides a strong foundation for understanding core programming concepts such as variables, loops, functions, arrays, and structures.

This practical project focuses on the development of a simple program using the C language. The project helps in improving logical thinking and problem-solving ability through hands-on implementation.

The main aim of this project is to apply theoretical knowledge of C programming into a working practical application.

02. Problem Statement

The problem addressed in this project is to design and implement a C program that performs a specific task efficiently.

The program is developed to solve a real-world problem using basic programming constructs such as input/output operations, conditional statements, loops, and functions.

The project demonstrates how C language can be used to create simple yet effective applications

03. Objectives of the Project

The objectives of this project are defined to provide a clear direction for the design and development of theThe objectives of this practical project are:

- To develop a working program using the C programming language.

- To understand the use of loops, conditions, and functions in problem-solving.

- To enhance programming and debugging skills.

- To implement a structured approach for writing efficient code.

- To gain practical exposure to real-time programming applications.

04. Tools and Technologies Used

The following tools and technologies were used in this project:

Programming Language: C

Compiler: GCC / Turbo C / Dev-C++

Development Environment: CodeBlocks / VS Code

Operating System: Windows / Linux

These tools provide a suitable environment for writing, compiling, and executing C programs.

05. Methodology

The methodology followed during the development of this project includes:

Understanding the Problem: Identifying the requirements and expected output.

Algorithm Design: Creating a step-by-step logical solution.

Program Development: Writing the code in C language.

Compilation and Execution: Running the program using a compiler.

Testing and Debugging: Checking for errors and improving accuracy.

Final Output: Ensuring the program produces correct results.

06. Results Analysis and Screenshots

This section describes the results obtained after implementing the C program.

The program was successfully executed and produced the expected output. It demonstrates the correct use of programming concepts and provides accurate results based on user input.

Output Screenshots:

(Insert program output screenshots here)

07. Logical Understanding

The logic of the program is based on:

- Taking input from the user
- Processing the data using conditions and loops
- Applying functions (if required) for modular coding
- Displaying the final output clearly

This project helps in understanding the flow of control and logical execution in C programming.

08. Conclusion

In conclusion, this practical project was successfully completed using the C programming language. It strengthened the understanding of fundamental concepts such as decision-making, iteration, and function implementation.

This project provides valuable experience in developing simple applications and improves overall programming confidence.

09. Future Scopes and Application

The future improvements and applications of this project include:

- Adding more features and functionality
- Improving efficiency using advanced concepts
- Extending the program for larger real-world applications
- Creating user-friendly interfaces
- Using file handling for storing and retrieving data

C programming projects are widely used in system programming, embedded systems, and software development.

References

- (1) Practical C Programming by B.M. Harwani (Packt)

Annexure

The following section includes the code of the game including the explanation for different portions.

<FileName >.cpp

```
#include "XXXX.h"
```

```
....
```

```
....
```

```
....
```

```
....
```

Main.cpp

```
#include "Game.h"
```

```
#include <ctime>
```

```
int main()
```

```
{
```

```
    srand(static_cast<unsigned>(time(nullptr)));
```

```
    Game game;
```

```
    game.run();
```

```
    return 0;
```

```
}
```